The multiplicity horn survives renormalization: exact quantum evolution of a ghost–normal field theory, the folded channel sum, and an accumulated-truncation floor

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Abstract

Two earlier papers of this program price the ghost-sector trilemma on a premise: that the quantum vacuum-decay rate of a coupled ghost sector scales with the number of open resonant channels—the *multiplicity horn*—so that the divergent field-theoretic rate is a channel-counting statement, not an artifact of perturbation theory. Against this stands the possibility that structural effects beyond perturbation theory—the position we test, framed as the Gross–Strumia–Teresi–Zirilli (GSTZ) metastability suggestion—could erode the channel sum. We put the premise to a falsification test in a genuinely quantum, multimode 1+1-dimensional two-field theory (normal φ plus ghost χ, contact coupling $\lambda\varphi^2\chi^2$, equal masses: maximal resonance, with the four-particle $\varphi\varphi\chi\chi$ shell exactly degenerate with the vacuum) by exact sparse-propagator evolution and a static-E_0 Feshbach/Schur folding of the full Hamiltonian onto the resonant shell. The pre-registered fork read: (a) any deficit of the exact rate against the bare channel sum is absorbed by standard renormalization (dressing) of the states entering the golden rule, or (b) a structural suppression survives folding. The data return (a), to the $\sim$1% level (with a documented downward drift of the static-E_0 fold): across couplings where the *naive* channel sum misses by up to 31%, and across mode numbers up to $M = 16$ where the naive ratio slides to 0.774, the folded (Schur) prediction tracks the exact evolution at 0.988–1.000. By elimination—diagonal self-energies and vacuum-column vertex corrections each reproduce the naive ratio to the four digits resolved—the deficit is attributed to the shell–shell block: first-order splitting and coherent mixing of the degenerate shell, textbook degenerate-level renormalization, no new mechanism. A dimension ladder calibrated to the exact Hamiltonian to 5.7×10^{-16} reproduces the program's reduced-measure rungs in quantum dress: the exactly resonant $d = 1$ shell saturates with cutoff, the $d = 1$ golden-rule sum grows log-class, and the $d = 2$ sum grows as a power (fitted exponent 2.26 against the predicted ≈ 2; a short-lever cutoff sweep, a mechanism check rather than a scaling measurement). A matrix-product-state cross-check certifies the position-space theory against exact diagonalization at $L = 4$ (to 6×10^{-6}), runs in production at $L = 16$, and then meets an honest limit that we characterize rather than hide: the Trotter/truncation floor is non-monotonic in the step (0.337/0.057/0.231 at $dt = 0.04/0.02/0.01$, $D = 40$), and raising D from 40 to 64 moves the floor from 0.057 to 0.062—saturation, per the pre-registered decision rule. The volume-scaling extension is therefore left open at the MPS level; the verdict itself rests on the exact methods. Conformal-gravity pricing that leans on the multiplicity premise survives this test; positions that require a nonperturbative erosion of the channel sum do not receive support at any scale we can reach.

Note on provenance. This manuscript is an experiment in AI-conducted research as much as a physics paper. The project began when the author asked the large language model Claude Fable 5 (Anthropic): *"Is it not possible that the antilinear PT/time-reversal symmetry element could maybe predict indivisible non-Markovian processes?"* Everything that followed was performed by the model: the research program, the theorems and proofs, the numerical experiments, the literature search, and this text. The model was repeatedly prompted to attack its own results adversarially and to search for prior work. The author, a professional engineer rather than a physicist, directed and reviewed the sessions but makes no claim to comprehension of the findings, and is circulating the manuscript for independent review by professional physicists. Accordingly, nothing here should be taken on trust: every numerical claim is reproducible from the public scripts (see Code availability); the originating conversation, project briefing, and full research notebook are preserved in the program's hub repository, `https://github.com/embeddetech/ghosts-program`.

1 Introduction

This is the seventh paper of a program that reads quantum dynamics in Barandes' currency of indivisible stochastic processes on configurations [6, 1, 2, 3, 4, 5]. Two of its predecessors lean on one premise. The vacuum-decay analysis of the ghost sector [2] factorizes the golden-rule rate into a boost-orbit volume times a reduced integral whose scaling is set by channel counting; the conformal-gravity pricing of [4] then prices the Lorentz-violating-cutoff horn of the trilemma on that reduced measure. Both inherit the field-theoretic position that the divergent decay rate of a coupled ghost vacuum is a statement about the *multiplicity* of open resonant channels—what those papers call the multiplicity horn.

That premise has a published antagonist. Gross, Strumia, Teresi, and Zirilli, asking whether negative kinetic energy is even metastable, concede the perturbative rate is "naively infinite" while suggesting the true nonperturbative rate is unknown and possibly benign [7]; a recent line of work constructs interacting ghost systems whose structure arrests the cascade entirely [8, 9]. If some structural mechanism—beyond the textbook renormalization of the states entering the golden rule—eroded the channel sum in a genuinely quantum evolution, the premise under two papers of this program would fail, and with it their pricing of the trilemma.

This paper is the falsification test. We build the smallest genuinely quantum, multimode theory in which the question is sharp: a normal field φ and a ghost field χ in 1+1 dimensions, momentum modes on a circle, contact interaction $\lambda \int \varphi^2 \chi^2$, equal masses. The ghost sign makes the four-particle $\varphi\varphi\chi\chi$ states with balanced momenta *exactly degenerate* with the vacuum: the resonant shell of the golden rule is not an approximation here but an exact zero-energy subspace, and its population count is the channel multiplicity. The pre-registered fork (recorded in the program notebook before the decisive runs):

(a) the exact decay of the vacuum tracks the channel sum after standard renormalization of the states entering it—the multiplicity premise survives; or

(b) a deficit survives the folding—incipient structural suppression, and the premise fails.

The data return (a) at every scale we can reach (Sec. 3). The deficit that perturbation theory does show is attributed, by elimination, to first-order splitting and coherent mixing of the degenerate shell (Sec. 4)—renormalization, not suppression. A dimension ladder reproduces the reduced-measure rungs of [2] in quantum dress (Sec. 5). A matrix-product-state cross-check certifies the position-space theory against ED at $L = 4$ (production at $L = 16$) and then hits a truncation floor

that we measure and report as a methods limit rather than extrapolate through—an accumulated-truncation floor (Sec. 6).

Honest scope, stated at the outset: everything here is 1+1-dimensional (with a $d = 2$ spot check), at most $M = 16$ (32 modes per field), occupation caps to 10, and a static-energy Schur fold. What the test shows is that finite-channel quantum dynamics contains no mechanism eroding the channel sum beyond textbook renormalization *above the* $\sim$1% *level reached here*; the infinite-channel, fully resummed limit—the true boost-orbit volume—remains analytic territory, priced in [2] and not re-derived here.

2 The model and the channel census

On $2M$ momentum modes per field ($k_n = 2\pi n/\ell$, $n \in \{\pm 1, \ldots, \pm M\}$, zero mode dropped, on a circle of length ℓ; the small-M stage-3 checks, the Sec. 2 channel ratios, the $d = 2$ spot check, and the ladder calibration use $\ell = 8$, while the campaign, rate-window, born-vacuum, and dimension-ladder scans use $\ell = 16$), with equal masses ($m = 1$) and free Hamiltonian $H_0 = H_\varphi - H_\chi$ (the ghost enters with the Ostrogradski sign), the contact interaction $\lambda \int \varphi^2\chi^2$ couples the Fock vacuum to four-particle states $\varphi_{k_1}\varphi_{k_2}\chi_{k_3}\chi_{k_4}$ with $\sum k_i = 0$. At equal masses the subset with $\{k_1, k_2\} = \{-k_3, -k_4\}$ pairings has free energy exactly zero: the resonant shell. The initial state is the negative-norm-free Fock vacuum of the free theory, whose excitations carry negative energy—the arena of the GSTZ framing [7]; we note that the integral-of-motion construction of [8] is built on $(x^2 - y^2)$-type interactions and does not extend to a pure $\varphi^2\chi^2$ coupling (there the biquadratic enters only as an explicitly nonintegrable perturbation), so their mechanism does not protect this coupling.

The channel census (`scripts/t5_channel_ladder.py`) counts resonant channels 10/21/36/55 at $M = 2/3/4/5$ (the coupled resonant pairings, already including the same-mode pairs; the full zero-energy subspace P quoted in Table 1 is larger, since it also collects the zero-energy two-particle $\varphi_k\chi_{-k}$ pairs—and, at larger particle-number caps, higher-number degenerate states—not only these four-particle pairings); channel couplings are $g = \lambda/(\ell\sqrt{w_1 w_2 w_3 w_4})$ (the w_i the four participating mode frequencies) with a $1/\sqrt{2}$ per identical pair, and the analytic channel table matches the exact Hamiltonian's vacuum column to 5.7×10^{-16} (`scripts/t5_dimension_ladder.py`)—the calibration that anchors the ladder to the same theory. The archived check is at $d = 1$, $\ell = 8$, $M = 4/6$, cap 4; the $d = 2$ rung applies the same box-normalized vertex without an archived exact-Hamiltonian comparison of its own.

Exact sparse-propagator evolution of $\langle N_\varphi \rangle(t)$ against the first-order channel sum gives ratios 0.999–0.995 out to $t = 20$ at $\lambda = 0.02$: at weak coupling the naive multiplicity premise is already nearly exact. The question is what happens when it is not.

3 The decisive test: folding the full theory onto the shell

At larger coupling the naive channel sum overshoots the exact evolution. The decisive question is whether the deficit is *renormalization* (the states entering the golden rule are dressed) or *suppression* (something erodes the multiplicity itself). The instrument that separates them is a (static-E_0) Feshbach/Schur fold: with P the projector onto the vacuum plus the full degenerate shell and Q the rest,

$$H_{\text{eff}} \;=\; PHP \;-\; PHQ\,(QHQ - E_0)^{-1} QHP, \qquad E_0 = 0,$$

configuration	full/naive	full/Schur	
$M = 2/3/4$, cap 4, $\lambda = 0.02$, $t = 40$	—	1.0000 / 0.9997 / 0.9993	
$M = 2/3/4$, cap 6	—	0.9998 / 0.9984 / 0.9963	
$M = 4$, cap 4, $\lambda = 0.04$	0.9434	0.9972	
$M = 10$, cap 4, $\lambda = 0.02$	0.8959	0.9951	
$M = 12$, cap 4	—	0.9941	
$M = 16$, cap 4	0.7737	0.9924	($\lvert P\rvert = 561$, $\lvert Q\rvert = 15{,}416$)
$M = 6$, cap 6	—	0.9884	($\lvert Q\rvert = 22{,}056$)

Table 1: Exact evolution versus the naive channel sum and the Schur-folded channel sum (`scripts/t5_stage3.py`, `scripts/t5_campaign.py`). Where the naive sum misses by 23% ($M = 16$), the folded sum tracks the exact rate to 0.8%. The $M \leq 4$ rows are the $\ell = 8$ stage-3 checks and the $M \geq 6$ rows the $\ell = 16$ campaign ($m = 1$ throughout); the naive ratio is box-dependent at the $\sim$ 1–2 percentage-point level, but the Schur-folded band is robust to ℓ, so the verdict does not depend on the box.

evolved exactly within P. The fold keeps every channel—the multiplicity is untouched—while dressing the shell and the vacuum by the full Q-space to second order in the fold and to all orders in the Q-space dynamics. If the deficit is renormalization, the folded prediction tracks the exact evolution; if multiplicity erodes by a mechanism beyond the reach of the static second-order dressing, it cannot.

Table 1 is the verdict. Across the campaign the folded ratio sits in the band 0.988–1.000 while the naive ratio slides from 0.995 to 0.774. The λ-scaling of the naive deficit at $M = 3$, cap 6 (0.0057/0.0227/0.0883/0.3144 across $\lambda = 0.005 \to 0.04$, close to $\times 4$ per doubling) identifies its leading piece as λ^2—the $(\Delta E\, t)^2$ signature of an $O(\lambda)$ splitting of the exactly degenerate shell—with the sub-$\times 4$ growth at the top of the range marking a negative (saturating) higher-order (λ^4) correction; the folded column stays at 0.0001–0.0069 throughout, including where the naive sum misses by 31%. Occupation-cap convergence (caps 4/6/8/10 at $M = 2$, dimensions to 2568) holds the folded ratio flat at 0.9998–1.0000 while the naive ratio settles non-monotonically near $\sim$0.956: the deficit is not a truncation artifact, and neither is its absorption by the fold. A $d = 2$ spot check (the 48-channel shell $\{-1, 0, 1\}^2 \setminus 0$, cap 4) gives full/Schur = 1.0000: the same behavior holds in the one $d = 2$ shell we can reach.

One honest sentence the table earns: the folded ratio drifts mildly downward as the fold deepens (1.0000 at small M to 0.9884 at the largest $|Q|$ we can invert), which is consistent with the expected degradation of the *static-E_0* Schur approximation rather than a resurgent deficit (fork (b) is thereby excluded only above the $\sim$1% level at the deepest reachable fold)—but we state it rather than absorb it, and the full fold at $|Q| \approx 81{,}000$ (cap-6 $M = 8$) exceeded the memory available to this program; the record there is the elimination chain of Sec. 4, not a folded ratio.

4 What the deficit is: attribution by elimination

At $M = 8$, cap 6, the naive ratio is 0.8620. Two partial dressings were computed exactly (`scripts/t5_partial_folds.py`) and both *fail* to move it: folding only the diagonal self-energies reproduces the naive ratio exactly, and folding the vacuum-column vertex correction reproduces it exactly again (0.8620 both times). What remains—and what the full fold captures at reachable M—is the *shell–shell block*: the direct couplings among degenerate shell members, their Q-induced second-order corrections, and the vertex corrections to non-vacuum columns. Both partial dressings

retain the naive first-order *structure* (a channel-by-channel sum of $|g_f|^2$ terms with no shell–shell rotation), which is why neither can move the ratio. Because the free energies vanish on the shell, the shell–shell block splits it at first order in λ; entering the transition probability as $(\Delta E\, t)^2$, that splitting reproduces the λ^2 growth of the deficit measured in Sec. 3. No wave-function Z-factor is involved: the static fold is Hermitized and evolved unitarily inside P (`scripts/t5_stage3.py`), so it carries $Z \equiv 1$ by construction, and in the exact evolution the *total* norm leaked out of P at this configuration is 6.3×10^{-3} against a 13.8% naive deficit—over an order of magnitude too small to be the leading piece. The deficit is thus attributed, by elimination (each partial dressing individually null; a shell-shell-*only* fold was not separately computed, and the full fold at this configuration is memory-limited, so the attribution leans on the reachable-M folds of Table 1), to degenerate-shell splitting—dressed kinematics, not eroded multiplicity. (The notebook records a failed expectation here: the author predicted the vertex correction would carry the deficit at the $\sim$0.98 level; it carried none of it. The prediction and its failure are both logged, per the program's pre-registration discipline.)

Two further signatures close the dynamical story (`scripts/t5_stage4.py`). The two-component window for the vacuum-survival rate $\Gamma(t)$ (the log-derivative of the survival probability), $\Gamma(t) = a + bt$ at cap 4, returns $b = 4.28 \times 10^{-5}$ against $2S_{\text{exact}} = 4.53 \times 10^{-5}$ ($\sim$6%; $S_{\text{exact}} = \sum_{\text{shell}} |g|^2$, each exactly-resonant $E \approx 0$ channel's squared coupling—the matrix-element modulus squared computed in the script), with the constant term $a = 2.03 \times 10^{-4}$ sitting just above the golden-rule plateau (1.56–1.84×10^{-4}). And the born-vacuum signature—in a deliberately detuned twin ($m_\chi = 1.35$, where the pair-channel detunings die as $1/k$ in the UV so the energy–time window $|\Delta E| < 2\pi/t$ can close; the twin is detuned in the *spectrum* only—the vertex normalization retains the $m = 1$ mode functions, a smooth $O(1)$ reweighting immaterial to the window kinematics read off here): the transfer rate is *sustained then collapsing* while the number of channels inside that window falls ($471 \to 7$ across $t = 5 \to 120$)—the vacuum decays as a multichannel resonance while channels are open and stops when they close. Whether the rate itself grows logarithmically with time in the born-vacuum regime is not resolvable at $M = 8$ and we do not claim it; the sustained-then-collapse profile is the signature shown.

5 The dimension ladder, in quantum dress

The reduced-measure analysis of [2] assigns vacuum-decay integrals to rungs by spacetime dimension and vertex class. With the channel table calibrated to the exact Hamiltonian (5.7×10^{-16}, Sec. 2), the same rungs can be read quantum mechanically. Three pre-registered predictions, all confirmed (`scripts/t5_dimension_ladder.py`; the golden-rule sums use a Gaussian smoothing width $\eta = 0.1$, and the quoted exponents shift by $\sim$0.01–0.15 across $\eta = 0.05$–0.2 without changing any verdict): the exactly resonant $d = 1$ shell *saturates* with cutoff (P1: +1.5% over the last doubling at $\Lambda = 100$); the $d = 1$ golden-rule sum grows *log-class*, the channel-table image of Garriga–Vilenkin's $D = 2$ logarithm [11] (P2: power-fit exponent 0.156 with still-decaying increments—consistent with a logarithm approaching its asymptote, and stated as log-class evidence rather than a converged asymptote); and the $d = 2$ sum grows as a *power* (P3: fitted exponent 2.263 against the predicted $\approx$2; the $d = 2$ scan is a short-lever cutoff sweep (cutoff $M = 4$–8, three-point fit), so this is a mechanism check, not a scaling measurement). Channel-table extensivity holds within $\sim$7.4% under volume doubling. (The notebook also records that the author initially mislabeled which dimension carries the log rung; the docstring was corrected against [2] before the first run.)

6 The MPS cross-check, and the accumulated-truncation floor

The exact methods above live in momentum space. As a fourth, independent cross-check we evolve the position-space lattice theory (two bosons per site, open boundaries, $H = H_\varphi - H_\chi + \lambda \sum_i \varphi_i^2 \chi_i^2$) by second-order TEBD, preparing the free vacuum as the imaginary-time ground state of the sign-flipped auxiliary Hamiltonian $H_\varphi + H_\chi$ (the same state; the ghost sign enters real-time evolution only), and reading the normal-sector energy transfer $\Delta E_\varphi(t)$ per interior site.

Certification. The gate run against dense exact diagonalization at $L = 4$ ($D = 100$, $dt = 0.02$, second-order Trotter, initialized from the exact ground state via dense diagonalization of the auxiliary Hamiltonian) agrees to 2–6 $\times 10^{-6}$ at $t = 1 \ldots 4$—up to three orders below signal (by $t = 4$; $\sim$50$\times$ below at $t = 1$). At $L = 16$, $D = 40$, $dt = 0.02$ (the production protocol): the $\lambda = 0$ floor is $\max|\Delta E| = 5.73 \times 10^{-2}$ (7.16×10^{-3} per interior site), the $\lambda = 0.06$ normal-sign twin sits at 5.41×10^{-2}, and the $\lambda = 0.06$ ghost run shows $\Delta E_\varphi/\text{site} = 8.93 \times 10^{-3}$ by $t = 12$—a signal 24.8% above the floor. At $D = 40$ only the $\lambda = 0.12$ rung separates cleanly from the floor ($\sim$1.9$\times$, on the diagnostic $dt = 0.01$ ladder described below); the $\lambda = 0.03$ rung is floor-buried, so the coupling-scaling claim of this paper is carried by the exact methods of Secs. 3 and 5, not by an MPS λ-ladder.

The floor, measured. The floor is not a precision issue: at 10^{-2}–10^{-1} it sits fourteen orders above machine epsilon, and it is non-monotonic in the Trotter step—0.337/0.057/0.231 at $dt =$ 0.04/0.02/0.01 ($D = 40$, $L = 16$)—the classic signature of a Trotter/truncation tradeoff: Trotter error falls as $T\,dt^2$ while truncation error, the Schmidt weight discarded at each re-compression to D, accumulates with the step count $\sim T/dt$. The sum is minimized near $dt = 0.02$, and refining the step *backfires* (a shared expectation of author and director, refuted by the data and logged). The mechanism is *not* a free-field entanglement barrier. At $\lambda = 0$ the free vacuum is an exact eigenstate of the real-time generator, and the $L = 4$ gate prepares it exactly (dense diagonalization of the auxiliary Hamiltonian); there its middle-cut entropy across the middle bond is stationary—0.2740 at every $t \in [0, 12]$—so there is no free-field entanglement growth to outrun the $S \sim \log D$ capacity. Nor is the floor static representability alone: at $L = 4$ the exact vacuum's discarded Schmidt weight beyond rank 40 at that bond is 2.2×10^{-11}, far below the observed floor. What accumulates is Trotter-injected excitation, repeatedly re-truncated: in the production protocol ($L = 16$, $D = 40$, $dt = 0.02$) the *simulated* state's middle-cut entropy climbs from 0.249 at $t = 0$ to 0.89 at $t = 12$, whereas at $L = 4$—the only size at which an exact reference exists—the exact state does not move and the $D = 40$ simulation tracks it to $\sim 10^{-3}$ (0.2740 $\to$ 0.2752). The $L = 16$ runs prepare the vacuum variationally, by annealed imaginary-time TEBD at $D = 40$, so their climb mixes Trotter-injected excitation with imperfect eigenstate preparation; the $L = 4$, $D = 100$ control (flat to 5×10^{-5}) is what shows the climb is not intrinsic. The pre-registered decision run ($D = 64$, $dt = 0.02$, $L = 16$) returned a floor of 6.22×10^{-2} against $D = 40$'s 5.73×10^{-2}: a 60% increase in bond dimension (a 13% increase in $\log D$ capacity) did not lower the floor (it rose slightly). Per the decision rule fixed before the run, we conclude the MPS is truncation-limited at this coupling and report it as a methods limit: *no volume-scaling claim is made from MPS data.* The diagnostic $dt = 0.01$ ladder (including an $L = 32$ point whose per-site transfer came out negative, i.e. non-extensive under truncation) is exactly that—diagnostic; volume extensivity at the field level is left open at the MPS level, supported here only by the exact channel-table extensivity ($\sim$7.4%) and the $L = 4$ ED gate test.

For an exact integrator none of this arises—which is precisely why the verdict of this paper rests on ED and the Schur fold, with the MPS supplying an independent $L = 4$ certification (production at $L = 16$) and an honestly characterized boundary of applicability.

7 Relation to prior work

Real-time false-vacuum decay on quantum lattices is an active lane: spin-chain false-vacuum decay [12], qumode MPS simulations of scalar-field decay—the nearest methodological template to Sec. 6 [13], and Fermi-golden-rule onset studies in discrete-mode systems [14]. On the ghost side, the state of the art in exact quantum treatments is the exact operator-conservation-law moment bounds of [9] and the integral-of-motion construction of [8]; the metastability suggestion tested here is that of [7], and the channel-counting frame descends from Cline–Jeon–Moore [10]; the low-dimensional log structure enters via the reduced-measure ladder of [2]. To our knowledge no published work evolves a multimode ghost–normal theory in real time, and none tests vacuum decay against channel count; both statements were re-checked against the 2025–2026 literature at drafting. The result is load-bearing in opposite directions for two camps: for the structural-stability program [8, 9] it marks where structure must enter (their mechanisms are interaction-structural, absent for $\varphi^2\chi^2$); for the metastability suggestion [7] it removes the only finite-channel mechanism we could construct.

8 Discussion

What is established, at what strength. *Exact*: the four-particle shell's exact degeneracy with the vacuum by construction, and the analytic channel-table calibration (matching the exact vacuum column to 5.7×10^{-16}). *Empirical (static-E_0 fold)*: the folded channel sum tracks the exact sparse-propagator evolution within the band 0.988–1.000 across every configuration where the fold is computable (Table 1), with the documented downward drift of the static-E_0 approximation; the two null partial folds of Sec. 4 (the shell-splitting attribution they support is inferred, not exact). *Computed, pre-registered*: the dimension-ladder rungs (P1 saturation; P2 log-class evidence, asymptote not pinned; P3 power, short-lever cutoff sweep). *Certified*: the MPS propagator against ED at $L = 4$, $D = 100$ (production at $L = 16$, $D = 40$), with a measured truncation floor beyond it. *Not claimed*: volume scaling from MPS; the born-vacuum log-growth of the rate; the infinite-channel resummed limit, which remains the analytic content of [2].

For the program, the consequence is simple: the multiplicity premise under the vacuum-decay pricing of [2] and the conformal-gravity fork of [4] survives its own falsification test. A structural suppression of the channel sum, if one exists, must live where this test cannot reach—in the infinite-channel limit, in higher dimensions at depth, or in interaction structures engineered for stability [8, 9]—and not in the generic finite-channel quantum dynamics of a coupled ghost.

Code availability. All numerical claims are reproducible from the public scripts in the program repository: `scripts/t5_channel_ladder.py`, `scripts/t5_stage2.py`, `scripts/t5_stage3.py`, `scripts/t5_stage4.py`, `scripts/t5_dimension_ladder.py`, `scripts/t5_campaign.py`, `scripts/t5_partial_folds.py` (Secs. 2–5); `scripts/t5_mps.py`, `scripts/t5_mps_gate.py`, `scripts/t5_mps_prod.py`, `scripts/t5_floor_diag.py`, `scripts/t5_run_one.py`, `scripts/t5_prod_dt01.py`, `scripts/t5_floor_mechanism.py` (Sec. 6; the last emits the entropy, Schmidt-weight and norm-leak diagnostics, and prints the entropy at *every* bond so the quoted middle-bond values carry their cut); a numerics-provenance map `scripts/SS5_NUMERICS.md`; result files (`results_val_M*_cap*.txt`, `results_partial_folds_M8_cap6.txt`, `mps_gate_result.txt`, `mps_prod_results.txt`, `dt01_*.txt`, `diag_*_c64_dt02.txt`) archived alongside.

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